Richitha Rekula

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Summary

Master of Science in Data Science, Analytics and Engineering candidate at Arizona State University, targeting data science, machine learning, and applied AI roles. Works on causal inference, forecasting, and ML systems, testing every result against a baseline and documenting its limits. Selected from 200+ applicants for a Zoom-sponsored fellowship building ZoomLens.

Education

Arizona State University — Tempe, AZ
Master of Science in Data Science, Analytics and Engineering | GPA: 3.78/4.0 — *Expected May 2027*

Anurag University — Hyderabad, India
Bachelor of Technology in Artificial Intelligence and Machine Learning | GPA: 3.53/4.0 — *Aug 2021 to May 2025*

Technical Skills

Languages: Python, SQL, Java, JavaScript, TypeScript, Rust, C++, C, R, MATLAB
AI & ML: TensorFlow, Scikit-learn, XGBoost, SHAP, NLTK, VADER, OpenCV
APIs & SDKs: Gemini API, Claude API, Zoom Apps SDK, Zoom Realtime Media Streams API, REST APIs, GraphQL
Data & Databases: PostgreSQL, MongoDB, Neo4j, SQLite, PySpark
Frameworks & Tools: Flask, Node.js, Express.js, SvelteKit, Streamlit, WebSockets, Docker, GitHub Actions, ServiceNow

Experience

Zoom Fellow — Tempe, AZ
ASU Next Lab, in partnership with Zoom — *Aug 2026 to Present*

- Built **ZoomLens**, a prototype AI assistant inside Zoom meetings that privately describes and explains shared-screen content for one participant at a time, without interrupting the presenter.
- Wrote the **Node.js** WebSocket server behind the **Zoom Apps SDK** panel, binding each connection to one participant with no broadcast path; verified by **7 isolation tests**, including 10 participants asking at once.
- Captured screens from the **Realtime Media Streams** feed, decoding a still with **ffmpeg** only on request (**25–110ms**).

Deskside Support Intern — Mesa, AZ
ASU Media and Immersive eXperience (MIX) Center — *May 2026 to Aug 2026*

- Diagnosed and resolved hardware, software, network, and account issues across classrooms and offices (full-time).
- Tracked incidents and documented resolutions in **ServiceNow**; troubleshot recurring user and system issues with IT.

Research Assistant — Hyderabad, India
Anurag University — *Aug 2023 to May 2025*

- Validated structured datasets and academic records across **5+ departments**, keeping data consistent for reporting.
- Reorganized digital filing and retrieval processes, cutting information access time by **20%**.

Projects

CauseWay: Causal Inference Benchmark | *Python, statsmodels, EconML, DoubleML* — 2026

- Estimated the effect of a promotional e-mail from a **64,000-customer** randomized experiment (**+$0.77** two-week spend per customer), with randomization tests, bootstrap intervals, and Holm correction.
- Removed randomization by rejection sampling and scored **7 estimators** (naive, IPW, matching, cross-fitted AIPW, Double ML) over **200 samples** per scenario: naive overstated the effect by **about 50%**, while IPW and AIPW landed within Monte Carlo error.
- With one confounder hidden, all methods stayed **14–47%** biased though refutation tests passed (E-values, Rosenbaum bounds).

GridCast: Day-Ahead Electricity Demand Forecasting | *Python, LightGBM, PyTorch, statsmodels* — 2026

- Backtested **6 models** (seasonal naive, harmonic regression with AR(1) errors, LightGBM, LSTM) on 24-hour forecasts of New England demand, using only data available at forecast time across **9 expanding validation folds**.
- Selected LightGBM on validation only; on a held-out 2024 test year it reached **705 MW MAE**, **22% below** the seasonal baseline, with 95% intervals covering **93.8%** of hours, falling to 84.5% during extreme demand.

Support Ticket Triage & Resolution-Time Predictor | *Python, XGBoost, PySpark, SHAP, Flask, Docker* — Jul 2026

- Built an ML pipeline on **20,000 synthetic** support tickets comparing Logistic Regression, Random Forest, and **XGBoost**; XGBoost reached **macro-F1 0.705** on priority and reduced resolution-time error to **12.2h** from a 24.4h baseline.
- Engineered text features with NLTK and VADER sentiment, added **SHAP** per-prediction explanations, and served the model through a Flask REST API (**15–40ms** responses) packaged with Docker.

Reddit Data ETL Pipeline | *PostgreSQL, SQL, Docker, GitHub Actions* — Feb 2026

- Built a **5-stage ETL pipeline** (extract, validate, stage, merge, check) loading **18.9M** Reddit records into PostgreSQL; `COPY` bulk staging and drop-and-rebuild constraints cut full-run time from 584s to **345s**.
- Enforced referential integrity at merge time, matching expected results exactly on a planted-defect test across all 18.9M records; wrote **49 automated tests** run on every push via GitHub Actions.